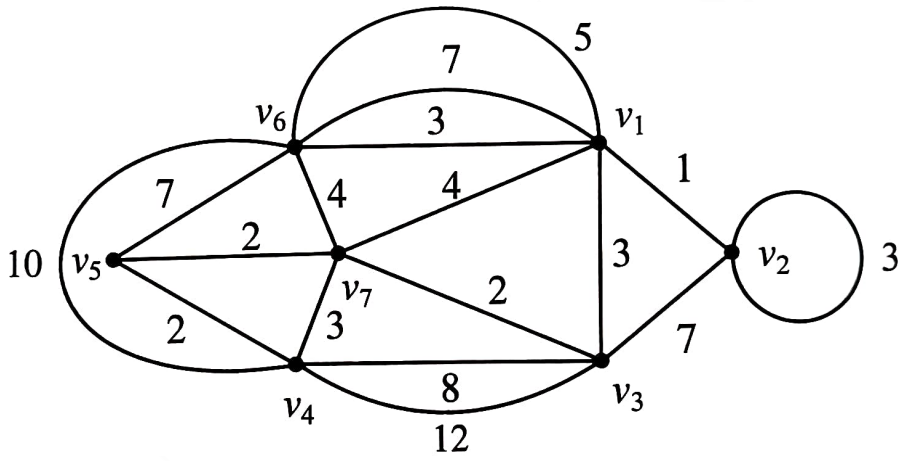
Dijkstra's Algorithm

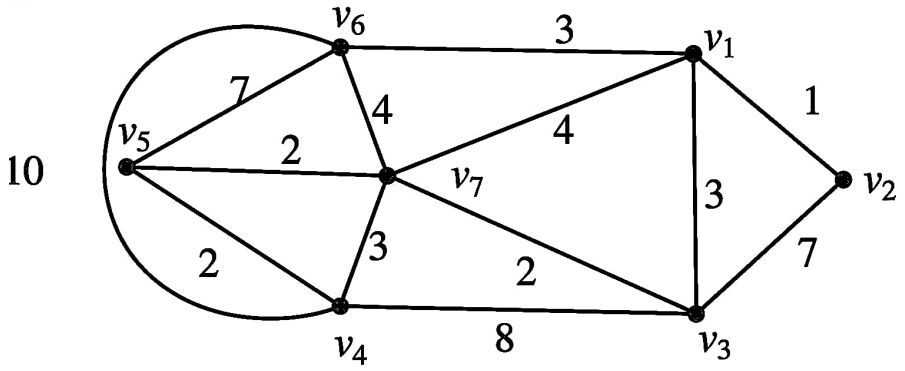
Dr. Sayantan Panja

Steps of Dijkstra's Algorithm:

Step 1: Initialization: (i) Assign distance 0 to the starting vertex S. (ii) Assign distance inf(infinity) to all other vertices. (iii) Mark all vertices as unvisited.
Step 2: Choose the Smallest Distance Vertex: Select the unvisited vertex with the smallest distance. Initially this is the source vertex S. Mark it as visited.
Step 3: Update Neighbour Distances: For each neighbour of the selected vertex: New distance= current distance+edge weight. If this new distance is smaller than the previously stored distance, update it.
Step 4: Repeat the Process:(i) Among all unvisited vertices, choose the one with the smallest distance. (ii) Mark it visited. (iii) Update the distances of its neighbours.
Step 5: Stop Condition: Repeat the process until the destination vertex is reached, or all vertices are visited.
Example: Using 'Dijkstra's algorithm' find the shortest distance from the vertex v_2 to v_5 in the following graph and hence find the shortest path:



Solution: The given graph has one loop (at v_2) and 3 parallel edges connecting v_1, v_6 and 2 parallel edges connecting v_5 and v_4. We discard the loop and the two parallel edges having weights 5 and 7(because the least weight among those parallels is 3). Similarly the edge v_3v_4 having weight 12 is deleted.



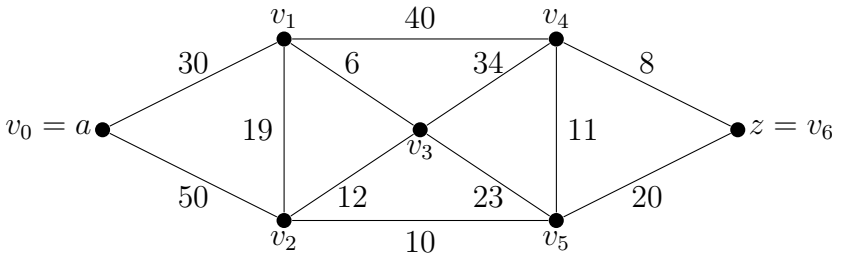
Here v_2 is the starting vertex and v_5 is the terminating vertex. The algorithm is given in the following table:

Dijkstra's Algorithm for finding shortest path from v_2 to v_5

	v_1	v_2	v_3	v_4	v_5	v_6	v_7	Process Described
S_1	∞	$\boxed{0}\checkmark$	∞	∞	∞	∞	∞	starting vertex v_2 is permanently labelled 0, others are temporarily labelled ∞.
S_2	$\min(\infty, 0+1) = \boxed{1}\checkmark$	$\boxed{0}$	$\min(\infty, 0+7) = 7$	$\min(\infty, 0+\infty) = \infty$	$\min(\infty, 0+\infty) = \infty$	$\min(\infty, 0+\infty) = \infty$	$\min(\infty, 0+\infty) = \infty$	v_1 is permanently labelled, as it is the minimum is this row.
S_3	$\boxed{1}$	$\boxed{0}$	$\min(7, 1+3) = \boxed{4}$	$\min(\infty, 1+\infty) = \infty$	$\min(\infty, 1+\infty) = \infty$	$\min(\infty, 1+3) = \boxed{4}\checkmark$	$\min(\infty, 1+4) = 5$	In this row, minimum is 4, which is a tie. In such case select any one of them.
S_4	$\boxed{1}$	$\boxed{0}$	$\min(4, 4+\infty) = \boxed{4}\checkmark$	$\min(\infty, 4+10) = 14$	$\min(\infty, 4+7) = 11$	$\boxed{4}$	$\min(5, 4+4) = 5$	In this row minimum is 4, so v_3 is permanently labelled.
S_5	$\boxed{1}$	$\boxed{0}$	$\boxed{4}$	$\min(14, 4+8) = 12$	$\min(11, 4+\infty) = 11$	$\boxed{4}$	$\min(5, 4+2) = \boxed{5}\checkmark$	v_7 is permanently labelled.
S_6	$\boxed{1}$	$\boxed{0}$	$\boxed{4}$	$\min(12, 5+3) = 8$	$\min(11, 5+2) = \boxed{7}\checkmark$	$\boxed{4}$	$\boxed{5}$	Stop the process as the destination vertex v_5 is permanently labelled.

Therefore, shortest distance from the vertex v_2 to v_5 is 7. Now, we will find the shortest path. Starting at the permanent label of v_5 (which is 7) we go back along the previously assigned temporary label of v_5, until we get a change. So we reach the temporary label 11 (at 5th row). In that row, permanent label vertex is v_7. Again we go back along the previously assigned temporary label of v_7, until we get a change. We reach the temporary label ∞ (at 2nd row). In that row permanent labelled vertex is v_1. Similarly, going back we reach the vertex v_2. Therefore, the shortest path is $v_2 \to v_1 \to v_7 \to v_5$.

Example: Find the shortest distance and hence the shortest path from the vertex a to the vertex z in the following graph. [Ans:77]



Example: Find the shortest distance from the vertex a to the vertex z and find the shortest path for the following graph. [Ans:16]

